

Project report

The Impact of Alcohol on Coding Skills

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Management Summary

1. Introduction

Nowadays, the IT sphere is becoming increasingly popular. High demand for IT specialties among applicants, as well as the variety of programs and courses attract more and more young students. Coding skills, along with other skills, are an integral part of the IT specialist profession.

In this article we will try to consider what impact alcohol has had on key skills for writing code

1.1 Context

While alcohol consumption is socially acceptable and common among students, many of them may not fully recognize the effects of alcohol on their ability to perform

technical tasks. We aim to understand potential consequences for cognitive performance in terms of writing code.

1.2 Main research question

The main research question guiding this project is:

"What is the impact of alcohol consumption on coding speed, productivity, and the number of errors made while programming?"

2. Methodology

Our study will use a combination of experimental data and survey to analyze the relationship between alcohol consumption and coding performance. We will collect both qualitative and quantitative data, this will ensure a more comprehensive approach. All participants will provide informed consent prior to the experiment, and their anonymity and confidentiality will be maintained throughout the study. To ensure safety, alcohol consumption will be carefully controlled, and participants' health and well-being will be monitored. They will be free to withdraw from the study at any time without consequence.

2.1 Research Strategy

- **Experiments:** Participants will perform coding tasks before and after consuming alcohol. Performance will be measured in terms of speed, productivity (lines of code written), and accuracy (number of errors). For this the LeetCode will be used.
- **Surveys:** We will collect information about participants such as: age, height, weight, drinking habits, skills and experience of coding.

2.2 Study Design

This study examines the effects of alcohol consumption on coding performance using structured experimental stages and accompanying survey data. Our analysis covers both quantitative measures of performance and subjective perceptions of participants.

1) Research Objectives and Sub-questions:

To explore alcohol's impact on coding, the study addresses the following sub-questions:

- I. **Sub-question 1: How does alcohol affect coding speed?**
 - Method: Participants will complete timed coding tasks both sober and post-alcohol. Times will be recorded and compared across different levels of alcohol consumption.
- II. **Sub-question 2: How does alcohol affect coding performance?**
 - Method: Performance will be measured by tracking the number of lines of code written and the difficulty of tasks completed.
- III. **Sub-question 3: What are the subjective perceptions of coding performance after drinking alcohol?**
 - Method: The survey will collect self-reports of participants' coding performance under the influence of alcohol, allowing comparison with objective experimental data.

2.3 Experiment set up

This section details the setup of experimental stages, measurement variables, and procedures used to obtain data across each alcohol consumption level.

1) Experimental stages

To assess the effects of different alcohol levels, participants will perform coding tasks in different experimental stages:

- **Stage 1:** [Survey](#) about participant coding skills, physical characteristics, and drinking habits
- **Stage 2:** Baseline (sober) - coding tasks are performed without drinking alcohol.
- **Stage 3:** After 1 drink - coding tasks are repeated after one standard drink.
- **Stage 4:** After 2 drinks - coding tasks are performed after the second drink.
- **Stage 5:** After 3 drinks - the final coding tasks are performed after three drinks

2) Variables

a. Dependent variables:

- Coding speed: measured by the time spent on coding tasks.
- Performance: measured by the total number of lines of code and the difficulty of the tasks completed.
- Error Rate: The number of syntactic and logical errors found in the code.
- Self-Perception: The participants' subjective assessment of their coding ability, collected via a post-experiment survey.

b. Independent Variable:

- **Alcohol Consumption Level:** Measured by the number of drinks consumed during the phase or by the corresponding estimated BAC levels.

2.4 Statistical Analysis Methods

To assess how alcohol affects coding performance, we will use the following statistical methods to allow for in-depth analysis of both performance measures and survey responses. We are limited to the small sample size so many of analysis methods are not usable for us.

Statistical Tests :

1. Paired t-Test

- a. **Objective:** To evaluate the average difference in coding performance (e.g., task completion time) between baseline (sober) and after each subsequent drink.
- b. **Use Case:** This test identifies statistically significant changes in participants' performance as alcohol consumption increases, offering insight into the impact of each additional drink on task efficiency.

2. Correlation Analysis

- a. **Objective:** To identify relationships between continuous variables like coding experience, drinking habits, and performance metrics.
- b. **Use Case:** Correlation analysis helps uncover meaningful links between participant characteristics (e.g., coding experience or drinking frequency) and coding performance, allowing us to determine if certain traits influence sensitivity to alcohol's effects on task speed.

3. Descriptive Trend Analysis

- a. **Objective:** To examine overall trends in coding performance across various levels of alcohol intake.
- b. **Use Case:** This trend analysis focused on visual patterns in coding speed across stages, providing a clear view of general performance shifts as alcohol levels increased, thus reinforcing broader conclusions about alcohol's impact on cognitive function.

3. Results and analysis

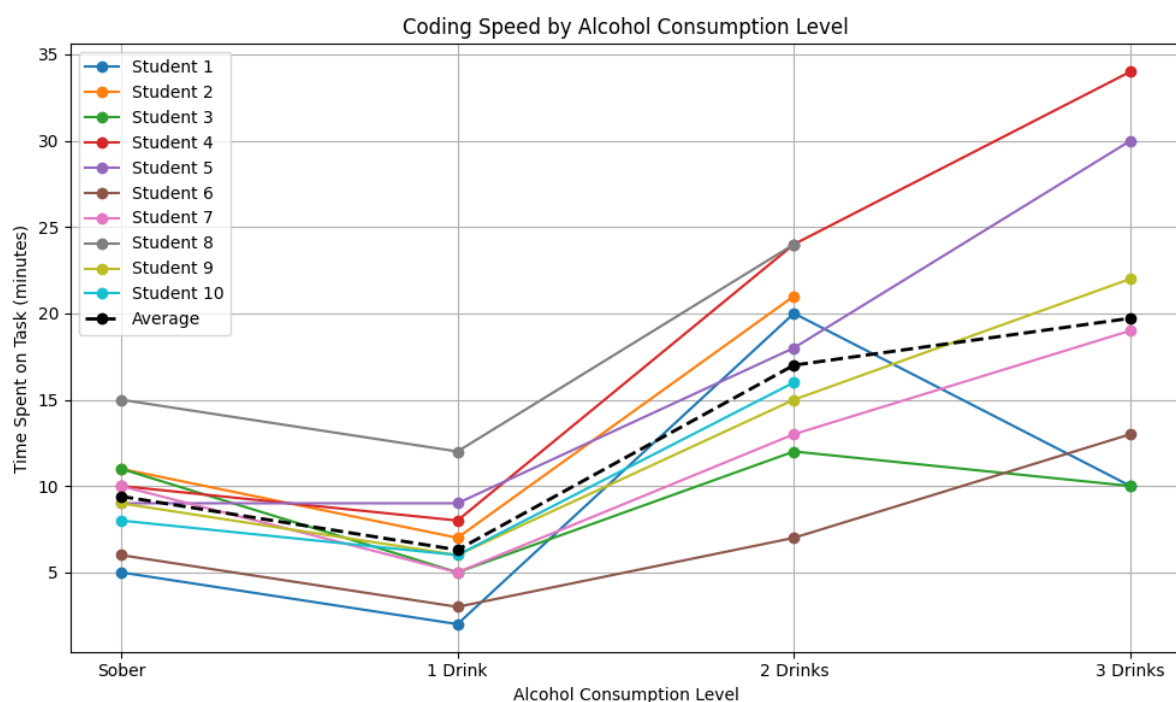
The experiment includes 10 participants, which is a relatively small sample size. We faced problems with recruiting, as some individuals were not willing to participate in our experiment. Additionally, two volunteers overestimated their coding skills, resulting in their inability to complete tasks even in the sober stage.

All analysis was done in Python. You can find this and Appendix 1.

3.1. How does alcohol affect coding speed.

This section examines the impact of alcohol consumption on the time that participants spent on coding tasks. Tasks were carefully chosen, so they are almost the same in terms of difficulty. However these tasks are on different themes, so that can play some role in the results and need to be taken into consideration as well.

1) Overall Coding Speed Trend.



Description: The first chart shows the overall trend for each participant and the average coding speed. As alcohol consumption increases, the time spent on tasks generally rises. Particularly after 2nd and 3rd drinks. Interestingly, we see a slight decrease in time after a first drink. Three participants haven't finished the last task.

Interpretation: This pattern highlights reduced brain efficiency with increasing alcohol consumption. The brief improvement after the first drink may reflect "The Ballmer Peak" effect. Effect that suggests that small alcohol intake can sharpen performance before

further consumption decreases it (Stone & Stoddart, 2024). However, given the small sample size, this effect cannot be confirmed in our study.

Also, as was mentioned earlier, a small variation in task difficulty can also play a role. Some participants may have more experience in some types of problems than in others, that may result in some participants solving the 3rd task faster than the 2nd one.

Paired t-Test Analysis for Coding Speed Across Alcohol Consumption Levels

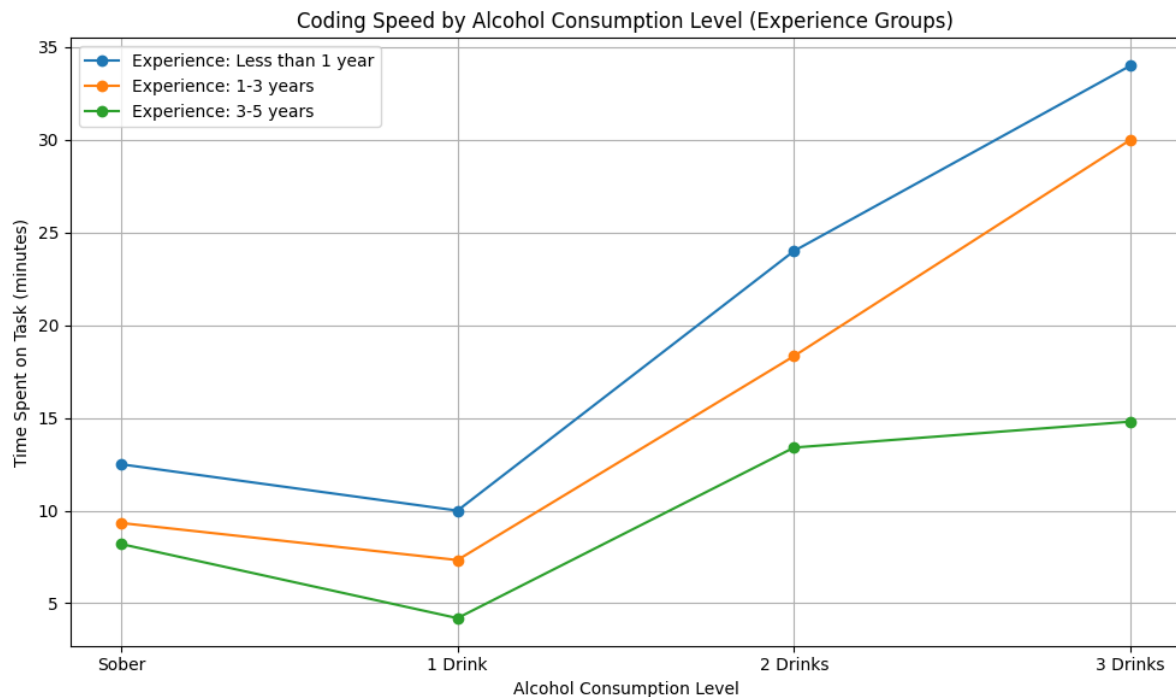
The paired T-Test was chosen to determine if there were significant differences between participant’s coding skills in different stages of the experiment.

Comparison	t-Statistic	p-Value	Interpretation
Sober vs 1 Drink	5.89	0.00023	Significant improvement in performance (time decreased) after 1 drink.
1 Drink vs 2 Drinks	-7.88	0.000025	Significant decrease in performance (time increased) after 2 drinks.
2 Drinks vs 3 Drinks	-1.44	0.20062	No statistically significant difference; performance remains similarly impaired after 3 drinks.

Description: The table displays the paired t-test results comparing coding performance between consecutive stages of alcohol consumption.

Interpretation: The improvement after one drink suggests a minor boost in performance. But after a second drink we see a significant decline in cognitive function. The difference between 2nd and 3rd drink is relatively small.

2) Experience Level Analysis



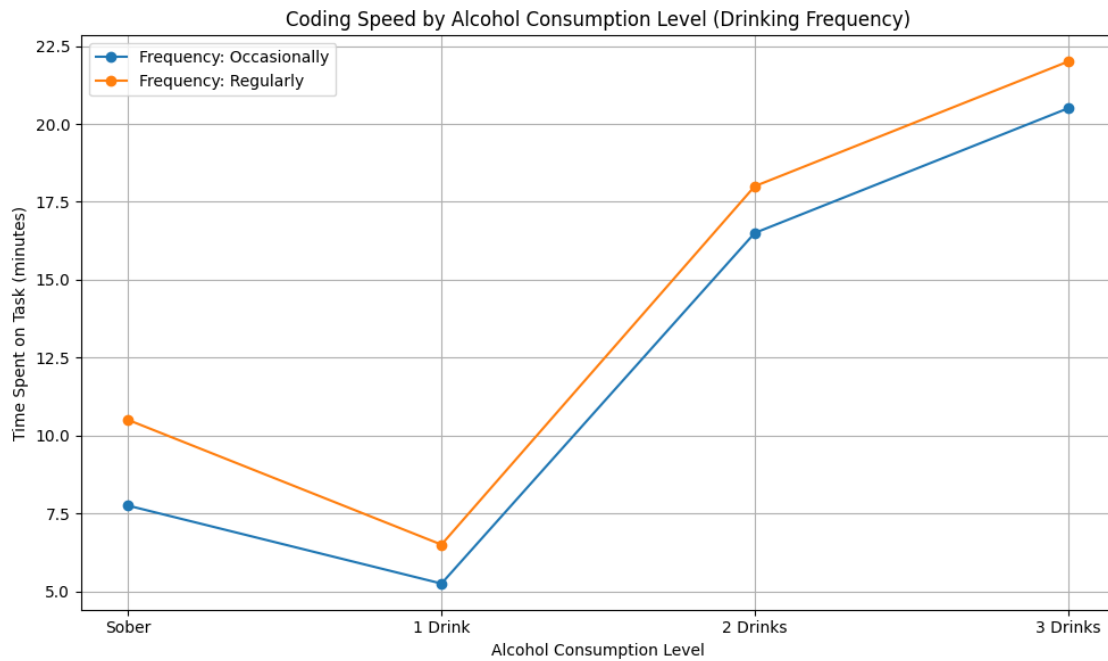
Description: The chart shows the overall trend in average coding speed for the participant grouped by skill in programming. Each line represents an experience group.

Interpretation: The trend suggests that experience level has an impact on alcohol tolerance and ability to perform coding skills. Coders with more coding experience appear to have a lower increase in task time. But the overall trend remains consistent, with every participant experiencing a decline in performance caused by alcohol.

Correlation Analysis: The correlation analysis further supports these observations. The results indicate a negative correlation in each steps. Negative correlation suggests that more experienced coders tend to solve problems faster than inexperienced ones, and are more resilient to alcohol.

- **Stage 'Sober':** Correlation = -0.59, p-value = 0.073
- **Stage '1 Drink':** Correlation = -0.83, p-value = 0.003
- **Stage '2 Drinks':** Correlation = -0.79, p-value = 0.007
- **Stage '3 Drinks':** Correlation = -0.86, p-value = 0.013

3) Drinking Frequency Analysis

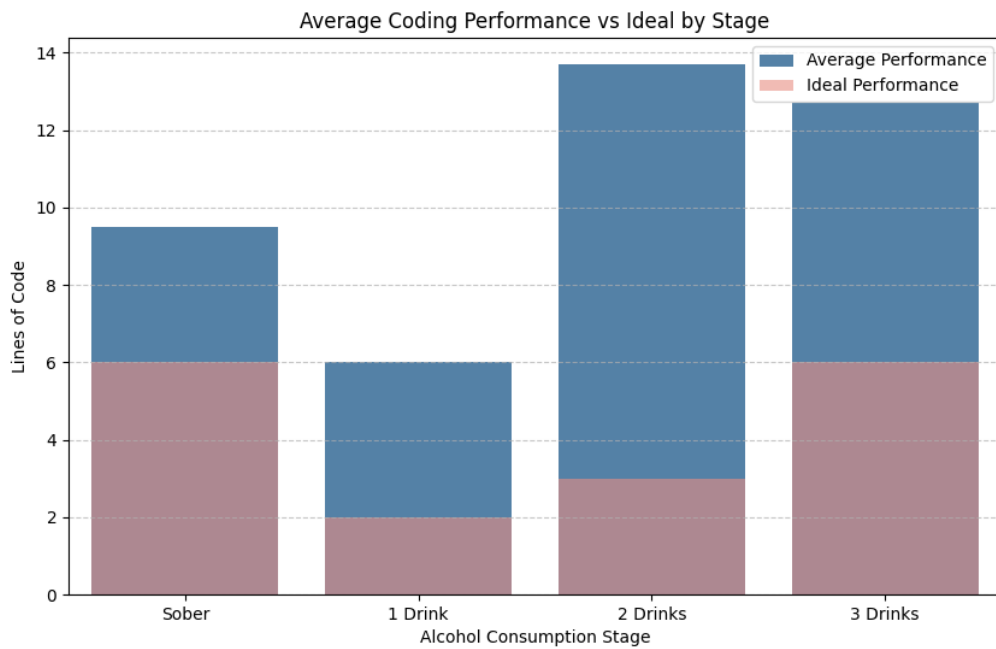


Description: This chart groups participants by weight and shows the average time spent on each task on different experiment stages.

Interpretation: While there is slight variation in performance between occasional and regular drinkers. The difference is minimal and do not point to anything.

3.2. How does alcohol affect coding performance.

1) Average coding performance vs ideal



Description: The bar chart titled "Average Coding Performance vs Ideal by Stage" displays the average number of lines of code written by participants at each stage of alcohol consumption: Sober, 1 Drink, 2 Drinks, and 3 Drinks. The chart includes two bars for each stage: blue bars representing the average number of lines of code (actual performance) and transparent salmon bars representing the "ideal" number of lines required for each task (based on LeetCode). The y-axis shows the average number of lines coded, while the x-axis represents the drinking stages.

In this context, closer alignment with the ideal (lower number of lines) represents better performance, as it indicates that participants were able to solve the tasks more efficiently.

From the chart:

- Sober stage: Performance is relatively close to the ideal number of lines, indicating efficient task completion.
- 1 Drink stage: This stage shows the higher deviation from the ideal, with participants producing more lines of code than necessary, indicating the lower performance.
- 2 Drinks stage: Participants wrote the highest number of lines of code in this stage, showing a significant drop in performance, as it is the farthest from the ideal.
- 3 Drinks stage: The average number of lines decreases slightly compared to the 2 Drinks stage, though the improvement is partly due to some participants being

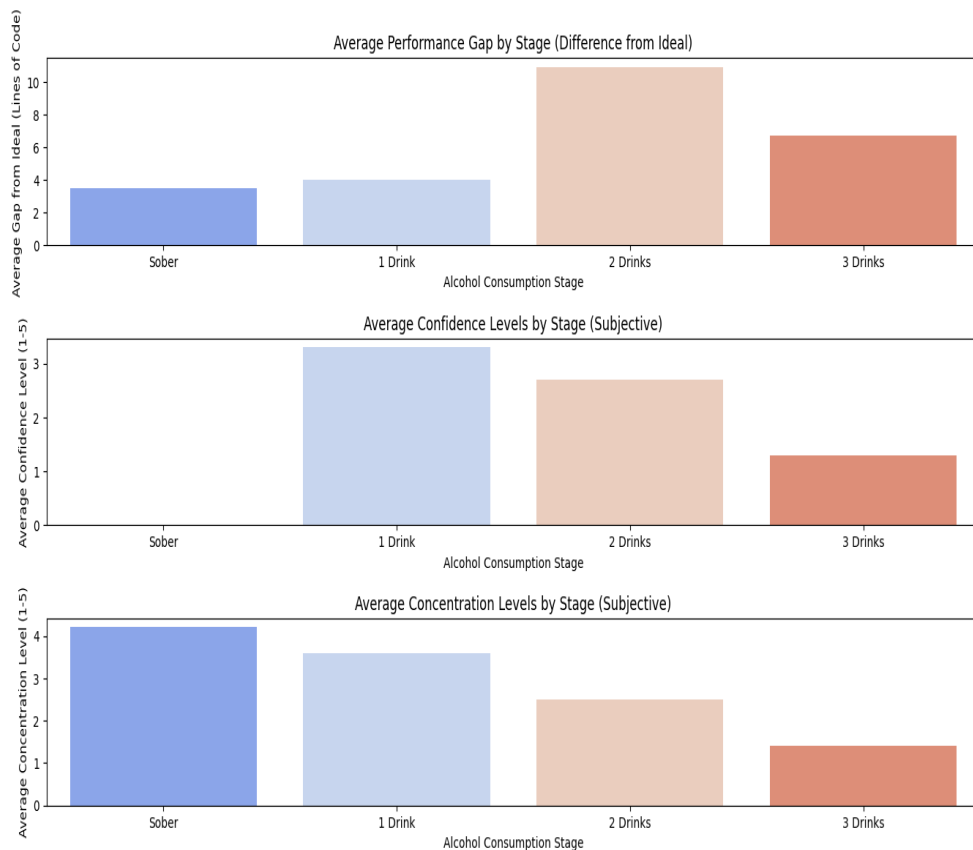
unable to complete the coding tasks at this level of intoxication, which skews the average downward.

Interpretation: The chart highlights a non-linear effect of alcohol consumption on coding efficiency, where higher numbers of lines coded indicate lower performance relative to the ideal.

- I. Initial Decline After One Drink: After consuming one drink, there is a notable drop in coding efficiency, as participants produced significantly more lines of code than ideal. This stage reflects the lowest performance, suggesting that even low levels of alcohol can interfere with cognitive functions, resulting in less efficient coding.
- II. Further Decline with Moderate Alcohol: At the 2 Drinks stage, participants wrote the most lines of code, deviating the farthest from the ideal performance. This stage indicates a further decline in coding efficiency, likely as alcohol increasingly impacts concentration and problem-solving skills.
- III. Apparent Improvement at Higher Consumption Levels: At the 3 Drinks stage, the average number of lines of code decreases slightly from the 2 Drinks stage. However, this apparent improvement is largely due to some participants being unable to write any code at this stage, resulting in an artificially lower average. This finding suggests that while moderate alcohol consumption impacts performance negatively, higher levels lead to such significant impairment that some individuals struggle to start or complete the tasks.

Overall Conclusion: The results suggest that any level of alcohol consumption is associated with reduced coding efficiency, with performance dropping further as consumption increases. While moderate consumption (2-3 drinks) shows the highest deviation from the ideal, even a single drink significantly impacts task efficiency. This analysis emphasizes that alcohol generally impairs coding abilities, with higher consumption causing the most pronounced decrease in performance.

2) Subjective assessments compared to actual



Description: These visualizations present the results of an experiment on coding performance at different stages of alcohol consumption: sober, 1 sip, 2 sips, and 3 sips. The impact of each stage is assessed both objectively and subjectively across three graphs:

- I. **Performance gap** (difference from ideal): The first bar graph shows the average deviation from the ideal number of lines of code (considered “ideal” for each stage). The y-axis represents the average gap in lines of code from this ideal, with lower values indicating better performance. The bars show that, on average, participants are furthest from the ideal number of lines at the 2 sip stage, suggesting that their coding performance deviates most from optimal under moderate alcohol influence.
- II. **Confidence levels** (self-reported): The second bar graph displays participants’ self-reported confidence in their coding performance at each stage, on a scale of 1–5. Lower values indicate lower confidence. This graph shows that participants' confidence gradually declines as alcohol

consumption increases, with the lowest confidence occurring at the 3-drink stage.

- III. **Concentration Levels** (self-reported): The third bar graph shows participants' self-reported concentration levels, also on a scale of 1-5. Like confidence, concentration declines with each stage of drinking, with the highest concentration levels occurring at the sobriety stage and the lowest at the 3-drink stage.

Interpretation: The results indicate a complex relationship between alcohol consumption, coding performance, and participants' perceptions of their abilities:

1. Objective Performance Gap:

- **Increasing Deviation from Ideal with Alcohol:** The gap from ideal performance increases as participants consume more alcohol, with the largest deviation occurring at the 2-drink stage. This suggests that moderate alcohol consumption results in the largest decline in coding performance, with participants producing many more lines than ideal, likely due to decreased concentration or difficulty refining their code.
- **Improvement at 3 drinks (due to non-completion):** Interestingly, the performance gap decreases slightly at 3 drinks. This may be partly explained by some participants not completing their tasks, resulting in a drop in line counts closer to the ideal. This does not indicate an improvement in performance, but rather reflects abandonment of the task due to severe impairment.

2. Self-reported confidence and concentration:

- **Confidence decline:** Participants' confidence in their coding results gradually declines with each additional drink. This decline coincides with an increasing gap from ideal performance, suggesting that participants may accurately perceive their decreased performance as their alcohol consumption increases.
- **Concentration decline:** Reported concentration levels also show a steady decline, paralleling the drop in confidence and performance bias. This suggests that participants' awareness of declining concentration may contribute to the decline in their confidence and observed performance.

3. Comparison of objective and subjective measures:

- Although both confidence and concentration levels decline with increasing alcohol consumption, the objective gap in performance shows that participants perform the least ideally at Stage 2 Drinks. This discrepancy may reflect a “false gain” effect, where participants are relaxed enough to produce more code but lack the precision needed to produce a quality result.
- By Stage 3 Drinks, both subjective and objective measures align more closely, as participants perceive difficulty in completing the tasks and rate their concentration and confidence as the lowest.

3.3 Other academic sources:

Author(s)	Findings
Balsa, Giuliano, & French (2011)	Alcohol increases consumed result in small but statistically significant cutbacks in GPA for male students, but not for females.
Ansari, Stock, & Mills (2013)	Alcohol misuse, particularly heavy episodic drinking, is highly likely to harm academic performance at university.
Bolin, Pate, & McClintock (2017)	Both alcohol and marijuana use predicted GPA; as students' alcohol and marijuana use increased, so did their GPAs.
Balsa, Gandelman, & Roldán (2018)	Test results proxy cognitive abilities; frequency of alcohol dependence in the last 30 days proxies risky behavior among 9 th and 10 th -grade students.
Sung, So, & Jeong (2016)	Negative association between alcohol consumption and academic achievement in adolescents, even after controlling for covariates.
Ukwayi, Lucy, Chibuzo, & Undelikwo (2013)	The majority of respondents are aged 15 to 35, which is associated with unhealthy behaviors characterized by substance use and abuse.
Onyebuchukwu, Sholarin, & Chico (2015)	Significant relationship between academic performance and alcohol

	consumption, with significant differences between drinkers and non-drinkers.
Meda et al. (2017)	Combined effects of alcohol and marijuana jeopardize college GPA, adding to teenage substance use literature.
Meier, Hill, Small, & Luthar (2015)	Adolescents from an upper-middle-class institution who used cannabis had lower GPAs and more externalizing symptoms in 12 th grade.
Tembo, Burns, & Kalembo (2017)	Students who consumed dangerous amounts of alcohol were 1.2 times more likely to report psychological distress than those who consumed less.
Ansari, Salam, & Suominen (2020)	Gender was associated with all alcohol behaviors after controlling for living situation, marital status, age, parental education, and year of study.
Welwel et al. (2022)	Many students have been involved in accidents and injuries due to alcoholism, resulting in illness and inability to attend lectures.
Evans-Whipp, Plenty, Catalano, Herrenkohl, & Toumbourou (2013)	Six students were excluded from the analysis; 44 others were removed due to honesty criteria.
Patte, Qian, & Leatherdale (2017)	Students who reported regular binge drinking were less likely to complete homework, attend class, value grades, or achieve high grades. Frequent binge drinking worsened outcomes.
Andacao & Lad (2021)	Students' overall level of alcohol consumption was moderate, and academic performance was satisfactory. A significant difference was found in college students' alcohol consumption.

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Research consistently shows a negative correlation between alcohol consumption and academic performance. Most studies highlight that even moderate alcohol consumption can harm cognitive function and lower GPA, with the effects often more pronounced among male students. Academic effects include absenteeism, decreased academic commitment, and increased risk-taking behavior. Some studies, such as

Bolin, Pate, and McClintock (2017), present results in which substance use is positively correlated with GPA; however, these are outliers in a largely negative picture.

Overall, the literature indicates that alcohol use is associated with lower academic performance and increased risk-taking behavior. Reducing alcohol consumption among students may lead to better academic outcomes, supporting the importance of targeted interventions in the educational environment.

4. Conclusion

Our study examined the impact of alcohol consumption on coding performance, focusing on key areas such as speed, efficiency, and error rate. Using experimental tasks and participant self-reports, we aimed to understand the nuances of how alcohol impacts coding skills at different levels of consumption. Here, we summarize the key findings and implications of our study.

1. **Coding speed:** The data indicate an overall decrease in coding speed as alcohol consumption increases. While there was a small improvement after one drink, likely due to a short-term relaxation effect often referred to as the “Balmer peak,” performance worsened with subsequent drinks. This trend was statistically significant, reinforcing the idea that alcohol reduces the cognitive efficiency required for complex tasks like coding.
2. **Work efficiency:** Alcohol consumption led to a significant decrease in coding efficiency, as evidenced by an increase in the number of lines coded relative to the ideal. Participants wrote more lines as they progressed through the stages of drinking, particularly at the two-drink stage, suggesting an impaired ability to focus and refine their work. However, after three drinks, there was a small decrease in the number of lines written, likely due to task abandonment due to severe cognitive impairment.
3. **Error rate and self-perception:** Error rate and subjective ratings reflected objective performance data. As participants consumed more alcohol, they reported decreased confidence and concentration, which corresponded to the observed increase in errors. This awareness of impairment suggests that

participants were able to recognize their own decline in ability, which could have implications for decision making under the influence of alcohol.

4. **Alcohol experience and habits:** More experienced coders showed slightly better tolerance to the effects of alcohol, but still experienced a decline in performance as consumption increased. The effect of alcohol frequency on coding performance was inconclusive, suggesting that individual variability in alcohol tolerance may play a role, but was not evident in this study.
5. **Comparison with literature:** Our results are consistent with existing research that generally links alcohol consumption to impaired cognitive function and academic performance. Research on student alcohol use consistently shows adverse effects on academic outcomes such as GPA, engagement, and task performance, highlighting the broader risks associated with alcohol use in educational settings.

The impact of alcohol on coding skills underscores the broader risks of cognitive impairment induced by alcohol consumption, particularly for tasks that require precision, problem-solving, and sustained focus. While there may be a slight initial benefit at very low levels of consumption, any further intake results in notable performance decline. Our research suggests that for students and professionals in fields requiring high cognitive function, even moderate alcohol consumption can undermine task efficiency and accuracy. Future efforts could benefit from exploring these effects across a larger sample size and varied task types. Reducing alcohol use among individuals in high-stakes technical environments could improve performance outcomes and enhance safety, reinforcing the importance of alcohol awareness and moderation initiatives in academic and professional spheres

References

1. Stone, T., & Stoddart, J. (2024). The Ballmer Peak: An empirical search. Proceedings of SIGBOVIK, Pittsburgh, PA, USA, April 2024 . <https://doi.org/10.48550/arXiv.2404.10002>
2. PP 77-84 ISSN: 2360-821X www.ajmrd.com

Appendix A

The data saved in the Google Sheets:

<https://docs.google.com/spreadsheets/d/1b5EF8hMXZeIRcnd58qt2aN0FUA6JxMrjqUb8-ov5hhk/edit?resourcekey=&gid=1603492440#gid=1603492440>

Link to repository with analysis code:

<https://github.com/TarasZinchenko/ProjectReportCode.git>